8.7 Practice: ready for the brugklas

← Unit overview · Answers

These are first looks at ideas you'll meet again in the brugklas. You don't need to be perfect at them. Draw a number line, a square or a machine whenever it helps. There is no clock.

Set A: Below zero (*negatieve getallen*) (Warm-up)

Use a thermometer or number line from −10 to 10.

1. It is 3 °C. The temperature drops 5 degrees. What is the temperature now?
2. $-2 + 6 = \ldots$
3. $5 - 8 = \ldots$
4. Put in order, smallest first: $-5 \cdot 2 \cdot -1 \cdot 0$
5. $-4 - 3 = \ldots$
6. In the morning it was −6 °C. In the afternoon it was 4 °C. How many degrees warmer did it get?
7. Femke has €20 on her bank account. She spends €25 (her bank allows it). What is her balance now?
8. A lift goes from floor 2 down to floor −1 (the car park). How many floors down did it go?
9. Which is warmer: −3 °C or −8 °C?
10. $-1 + 1 = \ldots$

Talk about it: For number 6, how did you count on the thermometer?

Set B: Squares and roots (*kwadraat en wortel*) (Building)

1. $3^2 = \ldots$
2. $5^2 = \ldots$
3. $10^2 = \ldots$
4. $1^2 = \ldots$
5. $\sqrt{16} = \ldots$
6. $\sqrt{49} = \ldots$
7. $\sqrt{100} = \ldots$
8. A square tile has an area of 36 cm². How long is each side?
9. $12^2 = \ldots$
10. $\sqrt{50}$ is between which two whole numbers?
11. 2^3 means $2 \times 2 \times 2$. What is 2^3?

Talk about it: Which square numbers do you just know? Which ones did you have to work out?

Set C: Number machines and formulas (Building → Stretch)

1. Machine: **in → × 2 → + 1 → out**. Put in 3. What comes out?
2. Same machine. Put in 5. Put in 10.
3. Machine: **out = in + 4**. Put in 6. What comes out?
4. Machine: **in → × 2 → + 1 → out**. 15 comes out. What went in? (Hint: run the machine backwards.)
5. Formula: *kosten* = *3* × *aantal* + *2*. What are the costs (*kosten*) if *aantal* = 4?

6. Look at the table.

in	1	2	3	4
out	5	8	11	14

   What is the rule? What comes out if you put in 10?

7. Bike hire: *prijs = 10 × dagen + 5* (€10 per day plus €5 for the helmet). What does 3 days cost?
8. If a = 4, work out: 3a (that means 3 × a), a + 7, and a^2.
9. A phone plan costs €8 per month plus €0,10 per text message. What do you pay in a month when you send 50 messages?
10. Machine: **in → × 5 → out**. Put in 0. What comes out?

Talk about it: In number 4, how do you run a machine backwards?

Set D: Which comes first? (*volgorde van bewerkingen*) (Stretch → finish on a win)

Remember: brackets first, then powers and roots, then × and : (left to right), then + and − (left to right). Put a box around the part you do first.

1. 2 + 3 × 4 = …
2. (2 + 3) × 4 = …
3. 10 − 6 : 2 = …
4. (10 − 6) : 2 = …
5. 3^2 + 1 = …
6. 20 : 4 × 5 = …
7. 18 − 4 − 3 = …
8. 5 × (8 − 3) = …
9. 2 × 3^2 = …
10. 100 − 10 × 3 + 5 = …
11. 1 + 1 × 1 = …

Talk about it: In number 6, why do we go left to right instead of doing × first?

Puzzle corner (optional, just for fun)

1. Add the odd numbers: 1 + 3 = …, 1 + 3 + 5 = …, 1 + 3 + 5 + 7 = … What do you notice? Can you draw why?
2. Put brackets in **4 + 4 × 4 − 4** to make the answer as big as possible. And as small as possible.
3. The machine **in → × 3 → − 6 → out** gives out the same number you put in. Which number?